13. The test statistic for testing the equality of two population means, when the population variances are known is

(a) $\dfrac{\overline{X}-\overline{Y}}{\frac{\sigma_X^2}{m}+\frac{\sigma_Y^2}{n}}$ (b) $\dfrac{\overline{X}-\overline{Y}}{\frac{s_X^2}{m}+\frac{s_Y^2}{n}}$

(c) $\dfrac{\overline{X}-\overline{Y}}{\sqrt{\frac{\sigma_X^2}{m}+\frac{\sigma_Y^2}{n}}}$ (d) $\dfrac{\overline{X}-\overline{Y}}{\sqrt{\frac{s_X^2}{m}+\frac{s_Y^2}{n}}}$

14. When the population variances are known and equal, the statistic $Z=\dfrac{\overline{X}-\overline{Y}}{\sigma\sqrt{\frac{1}{m}+\frac{1}{n}}}$ is used to test the null hypothesis

(a) $H_0 : \mu = \mu_0$ (b) $H_0 : \mu_X = \mu_Y$

(c) $H_0 : P_X = P_Y$ (d) $H_0 : P = p_0$

15. What is the standard error of the difference between two sample proportions, $(P_X - P_Y)$?

(a) $\sqrt{\hat{p}\hat{q}\left(\frac{1}{m}+\frac{1}{n}\right)}$ (b) $\sqrt{\hat{p}\hat{q}}\left(\frac{1}{m}+\frac{1}{n}\right)$

(c) $\hat{p}\hat{q}\sqrt{\left(\frac{1}{m}+\frac{1}{n}\right)}$ (d) $\hat{p}\hat{q}\left(\frac{1}{m}+\frac{1}{n}\right)$

16. What is level of significance?

(a) P(Type I Error) (b) P(Type II Error)

(c) upper limit of P(Type I Error) (d) upper limit of P(Type II Error)

17. Large sample test for testing population proportion is valid, when

(a) n is large, n ≥ 30, $np > 5$, $n(1 - p) > 5$ (b) n is large, n ≥ 30, $np > 5$, $n(1 - p) < 5$

(c) n is large, n ≥ 30, $np > 5$, $np\ (1 - p) > 5$ (d) n is large, n ≥ 30, $np > 5$, $np(1 - p) < 5$

18. Large sample test can be applied for two-sample problems, for testing the equality of two population means, when

(a) $m + n \geq 30$ (b) $m \geq 30$, $n \geq 30$

(c) $m \geq 30$, $m + n \geq 30$ (d) $mn \geq 30$

19. Large sample test is applicable, when the parent population is

(a) normal distribution only (b) binomial distribution

(c) Poisson distribution (d) any probability distribution

20. What is the rejection rule, based on large sample, for testing H_0 against one-sided (left) alternative hypothesis?

(a) $|z_0| \geq z_{\alpha/2}$ (b) $z_0 < -z_\alpha$

(c) $z_0 \leq -z_{\alpha/2}$ (d) $z_0 > z_\alpha$

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